Informatica®

Informatica® Cloud Data Integration

Amazon Athena Connector

Informatica Cloud Data Integration Amazon Athena Connector
April 2026

© Copyright Informatica LLC 2020, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Informatica, the Informatica logo, Informatica Cloud, and PowerCenter are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties. Required third party notices are included with the product.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-04-09

Table of Contents

Preface 5

Informatica Resources 5

- Informatica Documentation 5
- IDMC web site 5
- IDMC Communities 5
- IDMC Marketplace 5
- Data Integration connector documentation 6
- Informatica Knowledge Base 6
- IDMC Trust Center 6
- Informatica Global Customer Support 6

Chapter 1: Introduction to Amazon Athena Connector 7

Amazon Athena Connector assets 7

AWS Lake Formation use case 8

Chapter 2: Connections for Amazon Athena 9

Prepare for authentication 9

- Create an Amazon S3 policy 9
- Create an AWS Glue data catalog policy 10
- Create an Amazon Athena policy 11
- Configure EC2 Role to Assume Role 11
- AWS credential file authentication 13

Connect to Amazon Athena 13

- Before you begin 13
- Connection details 14
- Authentication types 14
- Advanced settings 17

Workgroups in Amazon Athena 17

Chapter 3: Mappings and mapping tasks with Amazon Athena Connector 18

Amazon Athena sources in mappings 19

Amazon Athena sources in mapping tasks 21

Rules and guidelines for mappings and mapping ctasks 21

Chapter 4: Migrating a mapping 22

Use the same object path for the migrated mapping 22

Use a different object path for the migrated mapping 22

- Migration options 23

Rules and guidelines for migrating a mapping 23

Chapter 5: Data type reference 25

Amazon Athena and transformation data types 25

Array, Map, and Struct data types 26

Chapter 6: Troubleshooting 27

Troubleshooting for Amazon Athena Connector 27

Index 28

Preface

Use *Amazon Athena Connector* to learn how to read from Amazon Athena by using Cloud Data Integration. Learn to create an Amazon Athena connection and develop and run mapping tasks and mappings in Cloud Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security policies and real-time system availability.

You can access the trust center at https://www.informatica.com/trust-center.html.

Subscribe to the IDMC Trust Center to receive upgrade, maintenance, and incident notifications. The Informatica Intelligent Data Management Cloud Status page displays the production status of all the Informatica cloud products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single component or all IDMC components. Subscribing to all components is the best way to be certain you never miss an update.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Amazon Athena Connector

You can use Amazon Athena Connector to connect to Amazon Athena to read flat files and Parquet files stored in Amazon S3.

Amazon Athena is a query service that you can use to analyze data in Amazon Simple Storage Service (Amazon S3) by using the standard SQL. Amazon Athena helps you analyze unstructured, semi-structured, and structured data stored in Amazon S3. Amazon Athena stores the schema in a data catalog or AWS Glue data catalog and uses it when you run queries. Amazon Athena uses Apache Hive to create tables and databases.

You can read data from views and external tables in the Athena data catalog and AWS Glue data catalog. Use Amazon Athena Connector to read and query Amazon S3 files that are mapped to the Amazon S3 location in an external table.

You can run queries in Amazon Athena on encrypted data in Amazon S3. You must run the Amazon Athena queries in the region where Amazon S3 is hosted. You can also encrypt the Amazon Athena query result stored on Amazon S3.

You can use workgroups in Amazon Athena to isolate and manage queries for different users, teams, or applications. You can configure each workgroup with specific settings, such as default S3 output locations and encryption options, to control how queries are processed and results are stored.

Amazon Athena Connector assets

Create assets in Data Integration to integrate data using Amazon Athena Connector.

When you use Amazon Athena Connector, you can include the following Data Integration assets:

- Data transfer task
- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

AWS Lake Formation use case

You can use Amazon S3 V2 Connector to write data to Lake Formation and use Amazon Athena Connector to read data from Lake Formation when Lake Formation is configured with specific permissions.

Lake Formation users can have specific permissions such as write access to Lake Formation, read with full access to external tables and columns, read with restricted access to external tables, and read with tag-based access to external tables.

For information on how to integrate data with Lake Formation, see Writing to and reading from AWS Lake Formation .

CHAPTER 2

Connections for Amazon Athena

You can use an Amazon Athena connection to read data from Amazon Athena.

Before you create a connection, enable the connector for the runtime environment. For more information about enabling connectors, see *Service and connector assignment* in the Administrator help.

You can use the connection to specify sources in mappings and mapping tasks.

Prepare for authentication

You can configure permanent IAM credentials, EC2 instance profile, EC2 role to asssume role, and AWS credential file authentication types to access Amazon Athena.

Before you configure the connection properties, create the minimal Amazon S3 policy, AWS Glue data catalog policy, and the Amazon Athena policies. Define the required permissions for the IAM user or EC2 role in the policies.

Attach the policies to the IAM user or EC2 role based on the authentication type that you want to configure.

Additionally, you need to keep the authentication details handy based on the authentication type that you want to use.

- To use the permanent IAM credentials authentication, create an IAM user, attach the required policies, and generate the access and secret key in the AWS Console. Keep these details handy to use in the connection properties.
- To use the EC2 instance profile authentication, install the Secure Agent on the EC2 instance and attach the EC2 role to the EC2 instance.
- To configure an EC2 role to assume another IAM role, install the Secure Agent on the EC2 instance, attach the EC2 role to the EC2 instance, and enable the EC2 role to assume another IAM role specified by the IAM Role ARN.
- To use the AWS credential file authentication, specify the profile name in the JDBC URL and ensure that the AWS credential file is available on the Secure Agent machine.

Create an Amazon S3 policy

Create an Amazon S3 policy in the AWS console and define the permissions to store Amazon Athena results on Amazon S3.

Use the following minimum required permissions to store Amazon Athena results on Amazon S3:

- PutObject

- GetObject
- DeleteObject
- ListBucket
- GetBucketLocation
- ListAllMyBuckets
- GetBucketAcl

You can use the following sample Amazon S3 policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Create an AWS Glue data catalog policy

You can use AWS IAM to define policies and roles to access resources used by AWS Glue.

Amazon Athena uses the AWS Glue Data Catalog to store and retrieve table metadata for the Amazon S3 data in your AWS account.

You can use the following sample policy for AWS Glue Data Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "glue:*",
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Create an Amazon Athena policy

Specify the minimum required permissions for Amazon Athena Connector to read data from views and external tables in the AWS Glue data catalog and to read and query Amazon S3 files.

You can use the following minimum required permissions:

- GetWorkGroup
- GetTableMetadata
- StartQueryExecution
- GetQueryResultsStream
- ListDatabases
- GetQueryExecution
- GetQueryResults
- GetDatabase
- ListTableMetadata
- GetDataCatalog
- CreatePreparedStatement
- DeletePreparedStatement

You can use the following sample policy for Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog"
                "athena:CreatePreparedStatement",
                "athena:DeletePreparedStatement"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:ListWorkGroups"
            ],
            "Resource": "*"
        }
    ]
}
```

Configure EC2 Role to Assume Role

You can configure an EC2 role to assume an IAM role and generate temporary security credentials to connect to Amazon Athena from the same or different AWS accounts.

The EC2 role can assume another IAM role from the same or different AWS account without requiring a permanent access key and secret key.

When you configure EC2 role to assume role, ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use the temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the EC2 role as a trusted entity allowing the EC2 role to use the temporary security credentials and access the AWS accounts.

When the trusted EC2 role requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted EC2 role.

Consider the following prerequisites before you use the EC2 role to assume role authentication:

- Install the Secure Agent on the AWS EC2 instance.
- The EC2 role attached to the AWS EC2 instance must not have access to Amazon Athena but must have permissions to assume another IAM role.
  The following is a sample trust policy that is attached to the EC2 role that allows the EC2 role to assume another IAM role:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "Statement1",
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::001234567890:role/aws_athena_assumeRole"
        }
    ]
}
```

- The IAM role that needs to be assumed by the EC2 role must have a permission policy and a trust policy attached to access Amazon Athena.
  The trust policy must contain the ARN of the EC2 role.

  You can also specify the external ID of your AWS account for a more secure access to Amazon Athena. The external ID must be a string.

  The following sample shows the assumed IAM role's trust policy with the ARN of the EC2 role and external ID:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001234567890:root" //anyone in this account
001234567890 can assume this role, this can also be limited to one role.
            },
            "Action": "sts:AssumeRole",
            "Condition": {
                "StringEquals": {
                    "sts:ExternalId": "athena_externalid"
                }
            }
        }
    ]
}
```

Note: When you configure an Amazon Athena connection, you must specify the ARN of the IAM role assumed by the EC2 role. Optionally, you can specify the External ID for a more secure access to Amazon Athena.

AWS credential file authentication

You can use a credential file that stores the AWS credentials, such as access keys and secret keys, to authenticate access to Amazon Athena.

Consider the following rules for an AWS credential file:

- The AWS credential file must be available in the following default location in your home directory on the Secure Agent machine:
  `~/.aws/credentials`
- If you want to use a different location for your AWS credential file, you can change the path by setting the environment variable named `AWS_SHARED_CREDENTIALS_FILE`. This variable should point to the full path of your AWS credential file.
  For example, `export AWS_SHARED_CREDENTIALS_FILE=/path/to/your/credentials`
- Ensure that the AWS credential file is syntactically correct. Incorrect syntax can cause connection failures.
  The following sample shows an AWS credential file:

```
[default]

aws_access_key_id = 1233333

aws_secret_access_key = abcabcabc


[test-profile]

aws_access_key_id = 1233333

aws_secret_access_key = abcabcabc
```

  The `aws_access_key_id` and `aws_secret_access_key` are the AWS access key and secret key used as part of credentials to authenticate the user.
- You can specify the profile name in the JDBC URL. If you do not specify the profile name, the credentials are used from the default profile in the AWS credential file.

Connect to Amazon Athena

Let's configure the Amazon Athena connection properties to connect to Amazon Athena.

Before you begin

Before you get started, you'll need to get information from your Amazon Athena account based on the authentication type that you want to configure.

To configure the permanent IAM credentials authentication, get the access key and secret key.

To configure the EC2 instance profile authentication, set up an EC2 instance and attach the EC2 role to the EC2 instance.

To configure the EC2 to assume another IAM role, set up an EC2 instance, attach the EC2 role to the EC2 instance, and enable the EC2 role to assume another IAM role specified by the IAM Role ARN.

To configure the AWS credential file authentication, specify the profile name in the JDBC URL and ensure that the AWS credential file is available on the Secure Agent machine.

Depending on the authentication method you choose, attach the appropriate policies to the IAM user or EC2 role.

Check out "Prepare for authentication" on page 9 to learn more about the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, serverless, or elastic runtime environment. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure permanent IAM credentials, EC2 instance profile, EC2 role to assume role, and AWS credential file authentication types to access Amazon Athena.

Permanent IAM credentials

Permanent IAM credentials authentication is the default type that requires the access key and secret key of the IAM user to connect to Amazon Athena.

The following table describes the basic connection properties for permanent IAM credentials authentication:

Property	Description
Access Key	The access key of the IAM user to connect to Amazon Athena.
Secret Key	The secret key of the IAM user to connect to Amazon Athena.
JDBC URL	The URL to connect to Amazon Athena. Enter the JDBC URL in the following format: `jdbc:athena://Region=<region_name;OutputLocation=<S3_Output_Location>;Workgroup=<Workgroup_Name>;` It is mandatory to specify the AWS region. You can either add the AWS region in the JDBC URL or specify it in one of the locations recognized by the AWS default region provider chain. For more information about the AWS default region provider chain, see the *AWS documentation*. If you use a workgroup with customer-managed query results, specify at least one of the two parameters in the JDBC URL, either the S3 output location or the workgroup name. For a workgroup with Athena-managed query results, specify only the workgroup name and do not include the S3 output location in the JDBC URL.

EC2 instance profile

You can configure AWS Identity and Access Management (IAM) authentication to connect to Amazon Athena when the Secure Agent is installed on an Amazon Elastic Compute Cloud (EC2) system.

The following table describes the basic connection properties for EC2 instance profile authentication:

Property	Description
JDBC URL	The URL of the Amazon Athena connection. Enter the JDBC URL in the following format: `jdbc:athena://Region=<region_name;OutputLocation=<S3_Output_Location>;Workgroup=<Workgroup_Name>;` It is mandatory to specify the AWS region. You can either add the AWS region in the JDBC URL or specify it in one of the locations recognized by the AWS default region provider chain. For more information about the AWS default region provider chain, see the *AWS documentation*. If you use a workgroup with customer-managed query results, specify at least one of the two parameters in the JDBC URL, either the S3 output location or the workgroup name. For a workgroup with Athena-managed query results, specify only the workgroup name and do not include the S3 output location in the JDBC URL.

EC2 Role to Assume Role

You can configure an EC2 role to assume an IAM role and generate temporary security credentials to connect to Amazon Athena.

The following table describes the basic connection properties for EC2 Role to Assume Role authentication:

Property	Description
JDBC URL	The URL to connect to Amazon Athena. Enter the JDBC URL in the following format: `jdbc:athena://Region=<region_name;OutputLocation=<S3_Output_Location>;Workgroup=<Workgroup_Name>;` It is mandatory to specify the AWS region. You can either add the AWS region in the JDBC URL or specify it in one of the locations recognized by the AWS default region provider chain. For more information about the AWS default region provider chain, see the *AWS documentation*. If you use a workgroup with customer-managed query results, specify at least one of the two parameters in the JDBC URL, either the S3 output location or the workgroup name. For a workgroup with Athena-managed query results, specify only the workgroup name and do not include the S3 output location in the JDBC URL.
IAM Role ARN	The Amazon Resource Name (ARN) of the AWS Identity and Access Management (IAM) role assumed by the EC2 role to use the dynamically generated temporary security credentials.

AWS credential file authentication

You can provide the AWS credentials in the credential file and place the file in the Secure Agent machine to authenticate access to Amazon Athena. When you configure an Amazon Athena connection, you can specify the profile name in the JDBC URL.

The following table describes the basic connection properties for AWS credential file authentication:

Property	Description

JDBC URL	The URL to connect to Amazon Athena.
	Enter the JDBC URL in the following format:
	`jdbc:athena://Region=<region_name;OutputLocation=<S3_Output_Location>;Workgroup=<Workgroup_Name>;`
	`ProfileName=<profile_name>;`
	It is mandatory to specify the AWS region. You can either add the AWS region in the JDBC URL or specify it in one of the locations recognized by the AWS default region provider chain.
	For more information about the AWS default region provider chain, see the *AWS documentation*.
	If you use a workgroup with customer-managed query results, specify at least one of the two parameters in the JDBC URL, either the S3 output location or the workgroup name. For a workgroup with Athena-managed query results, specify only the workgroup name and do not include the S3 output location in the JDBC URL.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Customer Master Key ID	The customer master key ID generated by AWS Key Management Service (AWS KMS) or the ARN of your custom key for cross-account access when you stage data in Amazon S3. The customer master key serves to encrypt your data at the destination before they are saved in Amazon S3.You can either enter the customer-generated customer master key ID or the default customer master key ID. Ensure that you generate the customer master key for the same region where your Amazon S3 bucket resides. For more information about using customer master keys with Amazon Athena, see Encryption in the AWS documentation.
External ID	The external ID associated with the IAM role. External ID is a unique, user-defined string value that the IAM role requires the EC2 role to provide when calling the sts:AssumeRole API.

Workgroups in Amazon Athena

You can use workgroups in Amazon Athena to isolate and manage queries for different users, teams, or applications. You can configure each workgroup with specific settings, such as default S3 output locations and encryption options, to control how queries are processed and results are stored.

When you create an Amazon Athena connection, you can specify a workgroup in the Amazon Athena JDBC URL

Consider the following guidelines when you use workgroups in Amazon Athena:

- When you use a workgroup with customer managed query results, you manage the storage for your queries and keep the results as long as you require.
  To use a workgroup with customer managed query results, you must specify at least one of the two parameters in the JDBC URL, either the S3 output location where you want to save the query results or the name of a workgroup with a preconfigured default S3 output location.

  To override the S3 output location and encryption settings specified in the JDBC URL, enable the **Override client-side settings** property when you create a workgroup with customer managed query results in Amazon Athena.
- When you use a workgroup with Athena managed query results, Amazon Athena manages the storage for your queries and retains query results for 24 hours.
  To use a workgroup with Athena managed query results, specify only the workgroup name and do not include the S3 output location in the JDBC URL.

CHAPTER 3

Mappings and mapping tasks with Amazon Athena Connector

Use the Data Integration Mapping Designer to create a mapping. When you create a mapping, you can configure a source to represent the object.

Amazon Athena source features

When you configure the advanced source properties, configure properties specific to Amazon Athena.

The following list shows the Amazon Athena source features that you can use in mappings:

- Server-side encryption
- Server-side encryption with KMS
- Client-side encryption with KMS
- Filter
- Sort
- Data Preview
- Table name override
- Schema name override

Data encryption in Amazon Athena sources

You can run queries in Amazon Athena on encrypted data in Amazon S3. You must run the Amazon Athena queries in the region where Amazon S3 is hosted. You can also encrypt the query results in Amazon S3.

Select the type of the encryption in the **Encryption Type** field under the Amazon Athena advanced source properties.

Use the customer master key ID generated by AWS Key Management Service for server-side encryption.

You can configure the following types of encryption:

None

The data is not encrypted.

SSE-S3

If you select the **SSE-S3** encryption type, Amazon Athena encrypts the file using the Amazon S3-managed key for server-side encryption.

SSE-KMS

If you select the **SSE-KMS** encryption type, Amazon Athena encrypts the file using the AWS KMS-managed key or Amazon Resource Name (ARN) for server-side encryption.

CSE-KMS

If you select the **CSE-KMS** encryption type, Amazon Athena encrypts the file using the AWS KMS-managed key for client-side encryption.

Amazon Athena sources in mappings

In a mapping, you can configure a Source transformation to represent an Amazon Athena source.

The following table describes the Amazon Athena source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. Select a source connection, or click **New Parameter** to define a parameter for the source connection.
Source type	Type of the source object. Select any of the following source object types: - Single Object. Select to specify a single Amazon Athena object. - Query. When you select the source type as query, you must map all the fields selected in the query in the **Field Mapping** tab. - Parameter. Select to specify a parameter name.
Object	Name of the source object. Select a single source object.

The following table describes the query options that you can configure in a Source transformation:

Property	Description
Filter	Filter value in a read operation. Click **Configure** to add conditions to filter records and reduce the number of rows that the Secure Agent reads from the source. You can specify the following filter conditions: - Not parameterized. Use a basic filter to specify the object, field, operator, and value to read specific records. - Completely parameterized. Use a parameter to represent the filter conditions. - Advanced. Use an advanced filter to define a complex filter condition. When you configure an advanced filter on a Timestamp column, you must specify the value as `TIMESTAMP` in the filter condition. For example, `SELECT * FROM <dbname>.<tablename> where col_date < TIMESTAMP '2030-06-22 18:30:00.000';`
Sort	Conditions to sort records. You can specify the following sort conditions: - Not parameterized. Select the fields and type of sorting to use. - Parameterized. Use a parameter to specify the sort condition. - Sort Order. Specify whether you want to sort data in ascending or descending order.

The following table describes the Amazon Athena advanced source properties that you can configure in a Source transformation:

Property	Description
Retain Athena Query Result On S3 File	Specifies whether you want to retain the Amazon Athena query result on the Amazon S3 file. Select the check box to retain the Amazon Athena query result on the Amazon S3 file. The Amazon Athena query result in stored in a CSV file. By default, the **Retain Athena Query Result On S3 File** check box is not selected. If you select this property for a workgroup with Athena managed query results, it is ignored.
S3OutputLocation	Specifies the location of the Amazon S3 file that stores the result of the Amazon Athena query. You can also specify the Amazon S3 file location in the `S3OutputLocation` parameter in the **JDBC URL** connection property. If you specify the Amazon S3 output location in both the connection and the advanced source properties, the Secure Agent uses the Amazon S3 output location specified in the advanced source properties.
Fetch Size	Determines the number of rows to read in one result set from Amazon Athena. Default is 10000.
Encryption Type	Encrypts the data in the Amazon S3 staging directory. You can select the following encryption types: - None - SSE-S3 - SSE-KMS - CSE-KMS Default is None.
Schema Name	Overrides the schema name of the source object. Maximum length is 128 characters.
Source Table Name	Overrides the table name used in the metadata import with the table name that you specify. Maximum length is 128 characters.
SQL Query	Overrides the default SQL query. Enclose column names in double quotes. The SQL query is case sensitive. Specify an SQL statement supported by the Amazon Athena database. When you specify the columns in the SQL query, ensure that the column name in the query matches the source column name in the mapping.

You can set the tracing level in the Amazon Athena advanced source properties to determine the amount of details that logs contain.

The following table describes the tracing levels that you can configure:

Tracing Level	Description
Terse	The Secure Agent logs initialization information, error messages, and notification of rejected data.
Normal	The Secure Agent logs initialization and status information, errors encountered, and skipped rows due to transformation row errors. Summarizes session results, but not at the level of individual rows.

Tracing Level	Description
Verbose Initialization	In addition to normal tracing, the Secure Agent logs additional initialization details, names of index and data files used, and detailed transformation statistics.
Verbose Data	In addition to verbose initialization tracing, the Secure Agent logs each row that passes into the mapping. Also notes where the Secure Agent truncates string data to fit the precision of a column and provides detailed transformation statistics. When you configure the tracing level to verbose data, the Secure Agent writes row data for all rows in a block when it processes a transformation.

Amazon Athena sources in mapping tasks

For Amazon Athena source connections used in template-based mapping tasks, you can configure advanced properties in the **Sources** page.

The **Sources** page appears in the mapping task wizard if you defined a parameter for the connection or source object in the associated mapping.

You can configure all the advanced source properties, except **Tracing Levels**, for a mapping task that you configured for a mapping. For more information, see “Amazon Athena sources in mappings” on page 19.

Note: If the mapping uses a specific source connection and a parameter for the source object, you can specify these advanced properties in the mapping and also specify them in the mapping task. In this case, the properties in the mapping task override those in the mapping.

Rules and guidelines for mappings and mapping ctasks

Consider the following rules and guidelines for mappings and mapping tasks:

- When you preview a column in a table that uses binary data type, the column displays blank values.
- The length of the schema name and table name must not exceed 128 characters at design time and run time.

CHAPTER 4

Migrating a mapping

If you configure a connection and mapping in one environment and then migrate and run the mapping in another environment, Data Integration uses the configured runtime attributes from the earlier environment and runs the mapping successfully in the new environment.

After the migration, you can change the connection properties from the Administrator service, but you do not need to modify the mapping.

Consider a scenario where you develop a mapping in the development organization (Org 1) and you then migrate and run the mapping in the production organization (Org 2). After you migrate, you might want to use the same or a different connection endpoint or object path in Org 2. Based on your requirement, follow the guidelines in this section before you plan the migration.

Use the same object path for the migrated mapping

If you want the migrated mapping in Org 2 to use the same object path as in Org 1, you must maintain the same schema and table in the Amazon Athena account for Org 2.

For example, if you have two different accounts, Account1 used for Org 1 and Account2 used for Org 2, the object path for the schema and table name must be the same in both the accounts:

Account1: AWSDATACATALOG1/SCHEMA1/TABLE1

Account2: AWSDATACATALOG1/SCHEMA1/TABLE1

In this scenario, you do not need to override the schema and table in the advanced properties.

Use a different object path for the migrated mapping

After you migrate the mapping, you can use a different object path to run the mapping from the new environment.

In this scenario, before you migrate the mapping, you can change the runtime attributes to reflect the object path in the migrated environment. You do not have to edit or update the mapping in the new environment.

As a rule, when you specify the schema and table in the advanced properties or object properties, Data Integration honors the attributes in the following order of precedence:

1. Runtime advanced attributes. The advanced properties such as the schema name and table name in the Source transformation in a mapping.

2. Object metadata. The object type selected in the Source transformation in a mapping.

Migration options

When you migrate, you can choose from one of the following options to update the object path:

Option 1. Update the connection properties to reference the new object

When you import the mapping into Org 2, in the **Review Connections** section, you can change the existing connection to map to the connection that has access to the specified schema and table in Org 2.

Option 2. Override the properties from the advanced properties

Before the migration, specify the required schema and table name for the object from Org 2 in the advanced properties of the Org 1 mapping.

After the migration, when you run the mapping, the Secure Agent uses the configured advanced parameters to override the object specified in the mapping imported from Org 1.

Option 3. Parameterize the properties in the mapping

You can choose to parameterize the advanced attributes, such as the schema and table name before the migration. You can configure input parameters, in-out parameters, and parameter files in the mapping.

After you migrate the mapping, do not edit or update the mapping. If you have used in-out parameters, you can change the schema and table attributes using the parameter file so that the changes are applied when the task runs.

Parameterizing only the advanced properties, but not the object in the mapping

If you want to parameterize only the advanced properties and use them at runtime, select a placeholder object in the object properties in the mapping and then specify an override to this placeholder object from the advanced properties. Ensure that the placeholder object contains the same metadata as the corresponding table that you specify as an override. When you run the mapping, the value specified in the advanced property overrides the placeholder object.

Parameterizing both the object and the advanced properties

If you want to keep both the Amazon Athena object type and the advanced fields parameterized, you must leave the **Allow parameter to be overridden at runtime** option unselected in the input parameter window while adding the parameters, and then select the required object at the task level. When you run the task, the values specified in the advanced properties take precedence.

Rules and guidelines for migrating a mapping

Consider the following rules and guidelines when you use the same or a different object path for the migrated mapping :

- The following table lists the transformation, object type, and the fields in the advanced properties of a mapping that you can retain when you migrate to the new environment:

Transformation	Object Type	Advanced Fields
Source	Single object	Schema name and table name

- Before you migrate a mapping to Org2, map the connection1 of Org1 to Org2 that can access the schema and table name configured in the advanced properties in Org1.
- After you migrate the mapping to Org2, you must not edit the mapping.

CHAPTER 5

Data type reference

Data Integration uses the following data types in mappings and mapping tasks with Amazon Athena:

Amazon Athena native data types

Amazon Athena data types appear in the **Fields** tab of the Source and Target transformations when you edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Amazon Athena and transformation data types

The following table lists the Amazon Athena native data types that Data Integration supports and the corresponding transformation data types:

Amazon Athena Data Type	Transformation Data Type	Description
Array	String	Maximum length is 65535 characters.
Binary	Binary	1 to 104,857,600 bytes You can use the Binary data type only with Parquet files.
Boolean	Integer	Logical Boolean (true/false)
Char	String	Fixed-length character string
Date	Date/Time	Calendar date (year, month, day)
Decimal	Decimal	Exact numeric of selectable precision
Double	Double	Precision 15
Float	Double	Precision 15

Amazon Athena Data Type	Transformation Data Type	Description
Int	Integer	Signed four-byte integer
Map	String	Maximum length is 65535 characters.
Smallint	Integer	Signed two-byte integer
String	String	-1 to 104,857,600 characters
Struct	String	Maximum length is 65535 characters.
Timestamp	Date/Time	Date and time (without time zone)
Tinyint	Integer	Signed one-byte integer
Varchar	String	Variable-length character string with a user-defined limit

For more information on how to read data of Array, Map, and Struct data types in JSON format, see "Array, Map, and Struct data types" on page 26.

Array, Map, and Struct data types

To read data of Array, Map, and Struct data types in JSON format, you need to set the `-DEnableHierarchicalDataParsingInAthena=true` property in the Secure Agent.

Perform the following steps to configure the Secure Agent:

1. Click **Administrator** > **Runtime Environments**.
2. On the **Runtime Environments** page, select the Secure Agent machine that runs the mapping.
3. Click **Edit**.
4. In the **System Configuration Details** section, select **Data Integration Server** as the **Service** and **DTM** as the **Type**.
5. Edit the JVM property and set the following value:
   `-DEnableHierarchicalDataParsingInAthena=true`
6. Click **Ok** to save the changes.

CHAPTER 6

Troubleshooting

Use the following sections to troubleshoot errors in Amazon Athena Connector.

Troubleshooting for Amazon Athena Connector

Unable to read unicode characters when using CSE-KMS encryption

When you read data that contains unicode characters, the characters are read as junk characters.

To resolve this issue, perform the following steps:

1. Click **Administrator** > **Runtime Environments**.
2. On the **Runtime Environments** page, select the Secure Agent machine that runs the mapping.
3. Click **Edit**.
4. In the **System Configuration Details** section, select **Data Integration Server** as the **Service** and **DTM** as the **Type**.
5. Edit the JVM Option system property and set the value to `-Dfile.encoding=UTF-8`.
6. Click **Ok** to save the changes.

Test connection fails with the Profile name not found error

The issue occurs when you initially specify a named profile and not the default profile in the JDBC URL. On the subsequent run, if you remove the profile name to let the driver use the default profile, the driver still looks into the cached credentials provider and tries to use the previously specified profile name from the first run. This behavior causes test connection failure.

To resolve the issue, you must set the profile name to default in the JDBC URL.

For example, `ProfileName=default`.

If the problem persists, spawn a new Data Integration Server.

INDEX

A

administration
  minimal Amazon IAM policy 9
Amazon Athena Connector
  assets 7
Amazon Athena sources
  mappings 19

C

Cloud Application Integration community
  URL 5
Cloud Developer community
  URL 5

D

Data Integration community
  URL 5
data type reference
  overview 25

I

Informatica Global Customer Support
  contact information 6
Informatica Intelligent Data Management Cloud
  web site 5

M

maintenance outages 6

S

status
  IDMC 6
system status 6

T

troubleshooting
  Amazon Redshift V2 Connector 27
trust site
  description 6

U

upgrade notifications 6

W

web site 5